

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May 2016

PH121 Pharmaceutical Biology-II

Date: 07.05.16 , Saturday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION – I

Q 1 Attempt all questions. Each question is of one mark. 20

1. T-Cell mediates
 - a) Cell mediated immunity
 - b) Humoral immunity
 - c) Non specific defence
 - d) None of above
2. Who is known as the father of genetics?
 - a) Mendel
 - b) Bateson
 - c) Watson
 - d) Morgan
3. Ascites are formed due to
 - a) Accumulation of fluid in the peritoneal cavity
 - b) Accumulation of fluid in the pleural cavity
 - c) Inflammation of lymph nodes
 - d) None of above
4. Point mutation involves
 - a) Change in single base pair
 - b) Deletion
 - c) Insertion
 - d) Duplication
5. The main extracellular cation is
 - a) Sodium
 - b) Potassium
 - c) Calcium
 - d) Magnesium

6. Replication is a process of
 - a) DNA duplication
 - b) m-RNA synthesis
 - c) Protein synthesis
 - d) None of above
7. Deficiency of which vitamin results in to Anaemia?
 - a) Vitamin B₁
 - b) Vitamin D
 - c) Vitamin E
 - d) Vitamin B₉
8. Lymph node present in the chest region is known as
 - a) Popliteal lymph node
 - b) Inguinal lymph node
 - c) Pelvic lymph node
 - d) Mediastinal lymph node
9. Vitamin D is synthesized with the help of which organ?
 - a) Skin
 - b) Liver
 - c) Kidney
 - d) All of Above
10. Which of the following enzyme is used to cut DNA molecule in rDNA technology?
 - a) Ligase
 - b) Phosphatase
 - c) Ribonuclease
 - d) Restriction enzymes
11. Who created first rDNA molecule?
 - a) Paul berg
 - b) Watson
 - c) Boyer and Cohen
 - d) Nathan
12. Which of the following is not relevant to lymph?
 - a) It is a tissue fluid
 - b) It supplies the nutrients where blood can't reach
 - c) It is a hormone
 - d) It contain lymphocytes
13. Anatomical barrier like skin is the type of
 - a) Adaptive immunity
 - b) Cell mediated immunity
 - c) Innate immunity
 - d) None of above
14. What is the most accurate statement about human pluripotent stem cells?
 - a) Can make all known types of human cells
 - b) Can differentiate and self-renew
 - c) Can make all types of human cells except extraembryonic tissues
 - d) Only embryonic stem cells fit that category

15. Which of the following is a water soluble vitamin?
 - a) Vitamin D
 - b) Vitamin E
 - c) Vitamin C
 - d) Vitamin K
16. What is true for bioremediation?
 - a) Purification of rDNA molecule
 - b) Use of organisms to neutralize pollutants from a contaminated site
 - c) Type of immunotherapy
 - d) None of above
17. Pandemic meaning
 - a) Regularly found among particular people or in a certain area
 - b) Prevalent over a whole country or the world
 - c) A widespread occurrence of an infectious disease in a community at a particular time
 - d) None of above
18. Following is not a type of immunotherapy
 - a) Monoclonal antibody
 - b) Antibiotics
 - c) Vaccines
 - d) Cytokines
19. Which vitamin helps in the process of blood coagulation?
 - a) Vitamin B1
 - b) Vitamin K
 - c) Vitamin A
 - d) Vitamin E
20. What percentage of body weight in men is water?
 - a) 20 %
 - b) 60 %
 - c) 30 %
 - d) 90 %

SECTION – II

- Q 2** Attempt any **FOUR** of the following
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|----------|---|-----------|
| A | Enlist the fat soluble vitamins and write a note on vitamin D. | 05 |
| B | Classify the total body water. Give a brief on intracellular fluid composition. | 05 |
| C | Describe the sources, recommended daily values and the uses of vitamin B complex. | 05 |
| D | Describe causative agent, mode of transmission and prevention of Malaria. | 05 |
| E | Explain the various mechanisms of water balance in human body. | 05 |
| F | Describe the composition and functions of lymph. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- A** Explain the application of biotechnology for gene therapy. 05
 - B** Explain the process of protein synthesis. 05
 - C** Describe in brief about various DNA repair mechanisms. 05
 - D** Write a note on cell mediated immune response. 05
 - E** Define Immunity. Write a note on natural immunity. 05
 - F** Draw a neat, clean and labelled diagram of lymph node and explain its histology. 05
- Q 4** Attempt any **FOUR** of the following
- A** Explain Bioremediation 05
 - B** Draw a neat, clean and labelled diagram of chromosome and explain its components. 05
 - C** Enlist the essential minerals present in our body. Describe in detail about anyone. 05
 - D** What is communicable disease? Describe causative agent, mode of transmission and prevention of Tuberculosis. 05
 - E** Explain application of transgenic animals. 05
 - F** Describe the histology and functions of spleen. 05
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